Driving to Work in Saudi Arabia

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Motivation

Gender and race-based barriers to labor market participation have long been associated with misallocation of talent and capital, with important impacts on aggregate productivity

- Hsieh et al. 2019: Reduced labor market barriers along racial and gender lines (labor market discrimination, barriers to human capital formation, and discriminatory social norms) can explain up to 40% of the growth in aggregate output between 1960 and 2010
- Chiplunkar and Goldberg 2024: Eliminating gender-related distortions suppressing female entrepreneurship and female labor force participation would yield higher real wages and profits, and reallocation of capital, generating substantial gains for the economy.

We examine the effect of a reduction in this type of barrier to women's employment in Saudi Arabia.

Literature

Reducing barriers to women's employment increases aggregate output:

- Hsieh et al (2019), Chiplunkar and Goldberg (2024), Ashraf et al. (2022), Bento (2020), Morazzoni and Sy (2022)

Transportation is particularly important for women's employment:

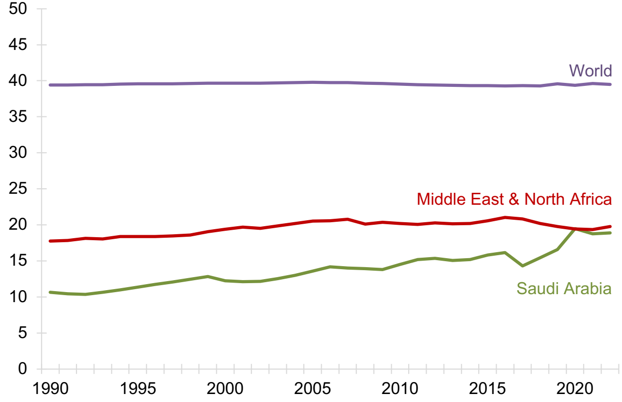
- Liu and Su (2020): Commuting time has higher disutility for female workers
- Abou Daher et al. (2023): Saudi women given assistance obtaining licenses were 61% more likely to drive and 35% more likely to be employed

Labor supply can react to anticipated future policy changes

- Jensen (2012), Blundell et al. (2015)

Women's Employment is Low in Saudi Arabia

Figure: Female Share of Labor Force



Source: World Bank

Saudi Women Face Significant Employment Barriers

- Social norms about gender-mixing
  - Strong social (and often legal) prohibitions against interaction and seclusion with non-related members of the opposite sex.
- Guardianship system

  Women required permission of a male guardian to work, travel and even obtain medical care.
- Transportation

  Women's driving ban, lack of public transportation
- Legal barriers
  - Many additional regulations around women's employment: hours, occupations, physical space, etc.

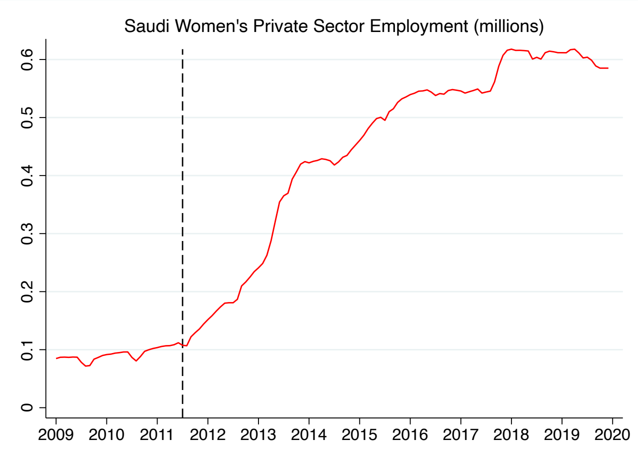
...But Things Have Been Changing Rapidly

All of this changed dramatically beginning in 2011, with a host of extremely ambitious post-Arab Spring labor reforms in the Kingdom.

- Private Sector Reforms (2011): Big push to open the private sector to Saudi citizens drastically increased the employment of *women* in particular
- Retail Sector Decree (2012): Required that *only* women could work in certain types of retail shops
- Lift of the Driving Ban (2018): Allowed women to drive motor vehicles
- Guardianship System Reforms (2019): Eliminated many guardianship requirements for adults.
- Women's Labor Reforms (2019): Eliminated sectoral restrictions, allowed night shift work, prohibited dismissal of pregnant workers, criminalized sexual harassment and mandated equal pay and retirement benefits.

Women's Employment Has Grown Significantly

Women' employment in the private sector has increased *over sixfold* since these changes began in 2011.



Labor Market Effects of Reforms

Previous work has explored the effects of some of these reforms on women's employment. In particular, the nationalization push yielded:

- Large increases in employment among women with secondary-level educations
- Integration of women into previously all-male firms
- Increase in Saudi employment (and increase in exit) at firms subject to nationalization quotas

This project looks at the effect of the *lift of the driving ban* on the labor market.

The Driving Ban

Saudi Arabia was previously the only country in the world where women were legally prohibited from driving motor vechicles

Longstanding protest movement began in the 1990s and ramped up in intensity after the Arab Spring uprisings in 2011

The government aggressively clamped down on these protests, arresting women who drove in protest of the ban.

لها حق القيادة

SHE HAS THE RIGHT TO DRIVE

#WOMEN2DRIVE

THE MIDDLE EAST CHANNEL

Saudi Arabia issues warning against women's driving campaign

Saudi Arabia has warned it may take action against women who participated in a campaign Saturday to defy a ban on women drivers. Women activists have been posting videos of themselves driving around the kingdom and claim they have an online petition demanding change with 16,600 signatures. This is the third effort of its kind, ...

The Driving Ban

The ban had been seen as a constraint to further employment growth for women and an economic problem for women and their families.

> *Transporting a woman takes two persons' time, and the time involved is significant taking into account the traffic congestion problems in most of the Saudi cities. Furthermore, in households without drivers, the husband is socially obliged to leave work to drive his wife if she needs to go to the doctor or other matters deemed important. Most employers, at least in the public sector, accept this cultural norm, implying that driving one's wife is a legitimate reason not to be present at work. (Hvidt, 2018)*

> *Lifting the driving ban would help new businesses, too. Saudi startup Yatooq adds the cost of a driver to salaries it pays women employees, and even places its factories in the city to make it easier for them to get to work.*
> *https://money.cnn.com/2016/11/30/news/economy/saudi-arabia-prince-women-drive/*

Lift of the Driving Ban

"Royal Order to Adopt the Provisions of the Traffic Law and Its Executive Regulation, Including the Issuance of Driving Licenses for Males and Females Alike"

- Lift of the driving ban was announced unexpectedly by King Salman on **September 26, 2017**
- The decree referenced the "negative effects of not allowing women to drive vehicles, and the positive effects envisaged from allowing them to do so"
- Press announcements emphasized that the ban had been a social issue, and that there was no actual law or religious edict prohibiting women's driving
- Women with a Saudi or GCC-issued driver's license were permitted to drive beginning **June 24th, 2018**

King Salman issues decree allowing women to drive in Saudi Arabia

King Salman. (SPA file photo)

Updated 27 September 2017

ARAB NEWS

September 27, 2017 01:07

JEDDAH: Saudi Arabia on Tuesday said it would allow women to drive in the Kingdom, in the latest move in a string of social and economic reforms underway in the country.

King Salman issued the decree, according to a royal court statement carried by the Saudi Press Agency (SPA).

Lift of the Driving Ban

Ford Middle East
@FordMiddleEast

Follow

Welcome to the driver's seat.
#SaudiWomenMove #SaudiWomenCanDrive

7:10 AM - 27 Sep 2017

Lift of the Driving Ban

The lift of the ban was hailed as an important step forward for women and their families

Marriam Mossalli, a Saudi entrepreneur ..., told CNN that "...The decision to let women drive will allow the debate to move on to other, more important issues, she said. "We can go now beyond that and look at the real issues we have, more entrepreneurs, more women in the workforce, and this is why the ban was lifted, to facilitate putting women in the workforce," she said. "A driver can be costly, around $400-800 a month, while an average entry level income for a woman working for example as a school teacher is $1,600 ... almost half of your salary is going to a driver. This is an economical decision and a human rights one."

https://www.cnn.com/2017/09/27/middleeast/saudi-arabia-women-drive

Research Questions

We think of the lift of the ban as removing a significant constraint on female workers, which we'd expect have positive effects on the allocation of *all* workers, with important productivity effects for firms

In particular, how did the lift of the ban affect:

- Overall employment
- Wage growth, occupation composition and job mobility
- Integration of previously all-male firms
- Did the effects correspond with the *announcement* of the change or the actual *implementation* of the lift?

We (plan to) use our modeling exercise to estimate:

- General equilibrium effects on aggregate productivity and output
- Potential for complementatry policies (e.g. lowering integration barriers) to increase the gains from reducing mobility barriers

Data

Job-level data from General Organization for Social Insurance (GOSI)

- Spell-level data on all private-sector firms that pay social insurance for Saudi employees
- Covers Saudi workers in private sector
- Monthly, covering 2009-2019
- Includes data on:
  - worker: gender, education, age
  - job: occupation, salary, start/end dates
  - firm: industry, location

Summary Statistics

	Jan 2017	Dec 2019
Number of Firms	240,407	182,439
Number of Workers	9,893,995	7,287,536
Number of Saudi Workers	1,825,120	1,827,946
Number of Male Saudi Workers	1,305,119	1,273,494
Number of Female Saudi Workers	520,001	554,452
Female Share of Saudis	28.5%	30.3%
Average Wage for Saudi Workers (SAR)	5082.73	5592.68
Average Wage for Male Saudi Workers (SAR)	5542.10	6110.83
Average Wage for Female Saudi Workers (SAR)	3929.80	4402.56
Share of Gender-Integrated Firms	48.6%	55.0%
Average Monthly EE transitions: Male	0.66%	0.58%
Average Monthly EE transitions: Female	0.66 %	0.72%
Average Monthly UE transitions: Male	1.60%	1.23%
Average Monthly UE transitions: Female	2.70%	2.37%

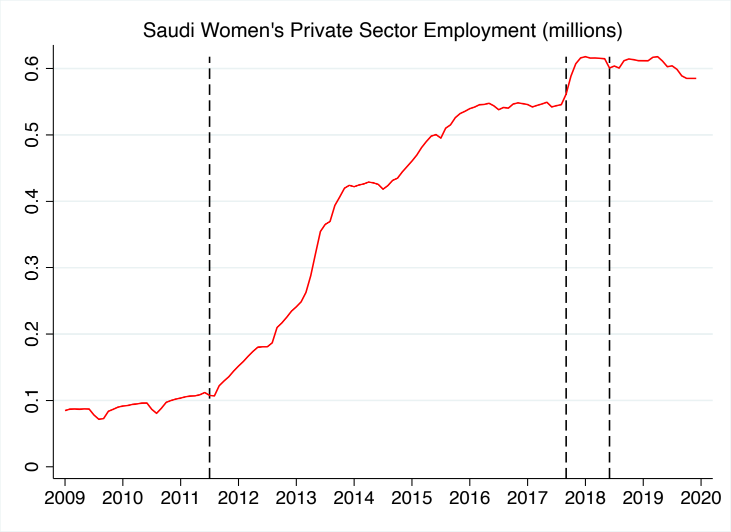
Empirical Patterns

- Employment jumps for women (and men), with a large incease in the female share of employees
- Large increase in the share of firms with any female employees ("gender-integrated" firms)
- Increases in job transitions generate wage growth and occupation advancement for employed workers
- New employment and job mobility yield sharp increases in the total wage bill for women and men after the announcement
- Women are more likely to join firms with (initially) fewer female employees, and men are more likely to join firms that had more female employees

The largest changes correspond with the *announcement* of the policy

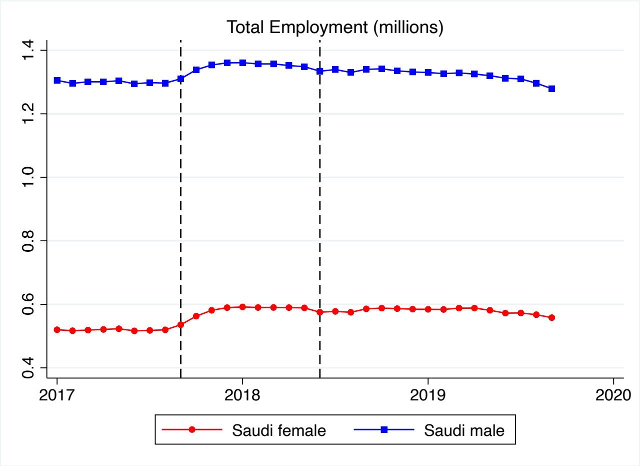
Women's Employment

The announcement of the policy corresponds with a jump in women's employment in the private sector:



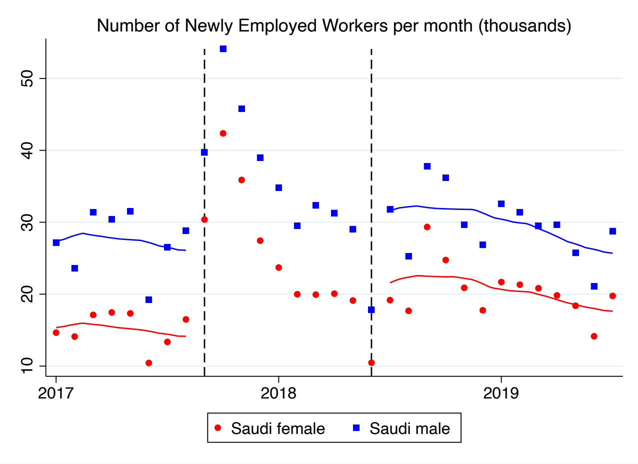
Men's Employment

Private sector employment also increased for Saudi men, but by a smaller percentage:



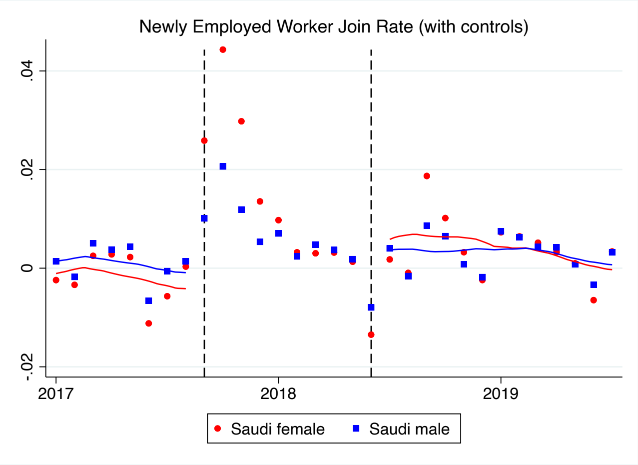
Labor Market Entry

There is a brief spike in entry of new workers corresponding with the timing of the policy announcement. Post-implementation entry levels remain higher than pre-announcement.



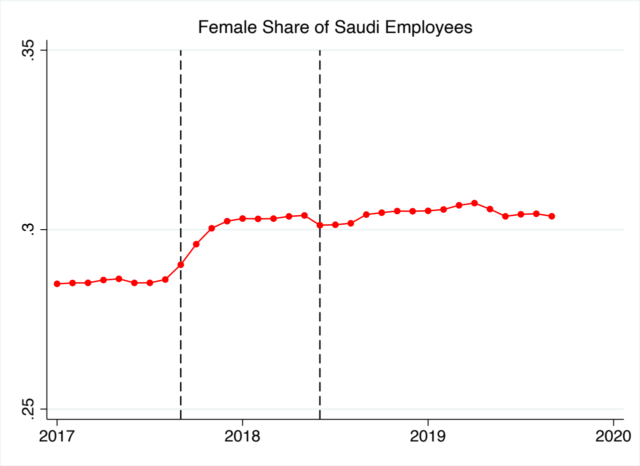
Labor Market Entry

Patterns for men and women are similar when controlling for age and education:



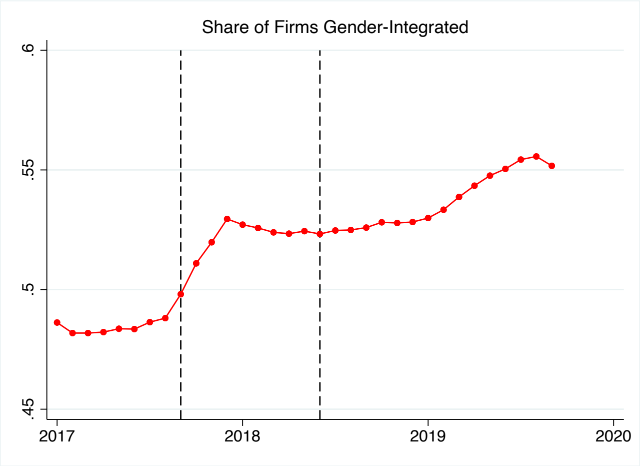
Female Share of Saudi Workers

Similar increases in employment yield a correponding jump in the female share of Saudi workers:



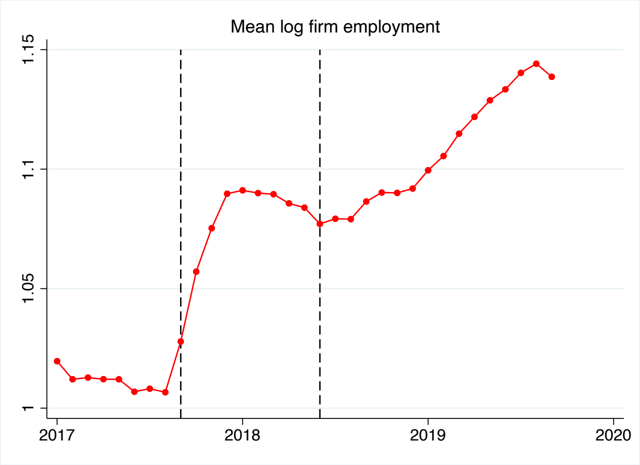
Firm Integration

There was a corresponding surge in the share of firms that had both male and female employees:



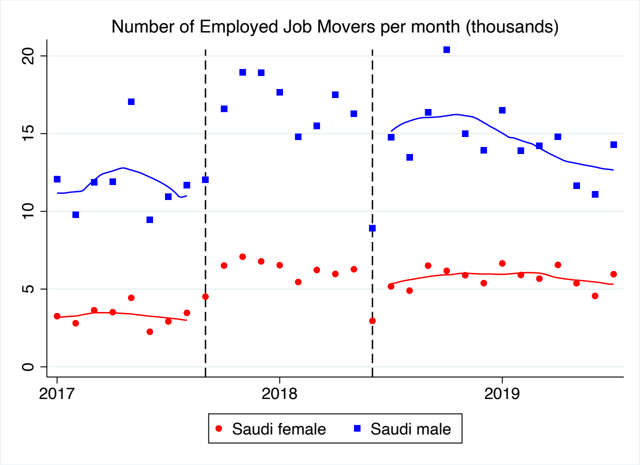
Firm Employment

Per-firm employment of Saudis also increased sharply with the announcement of the lift of the ban:



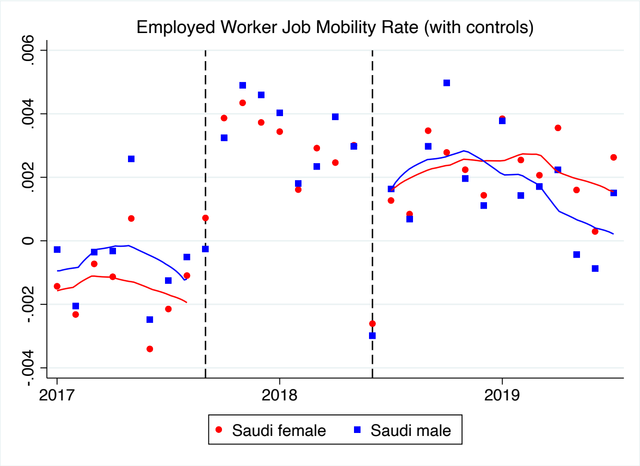
Job Mobility

Both men and women are more likely to switch jobs after the policy announcement



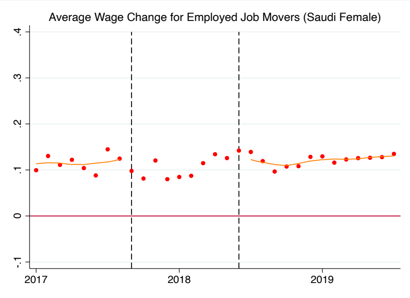
Job Mobility

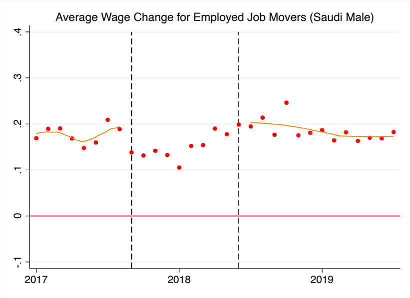
The jump in job transition rates is similar for men and women when controlling for age and education:



Wage Growth from Job Mobility

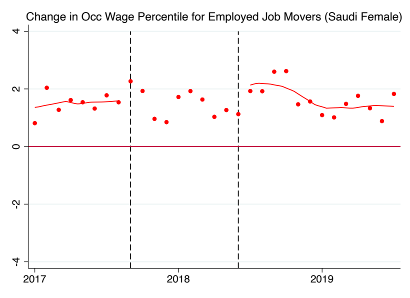
Wage growth for movers remains relatively stable, which implies that the increased mobility should yield higher average wages.

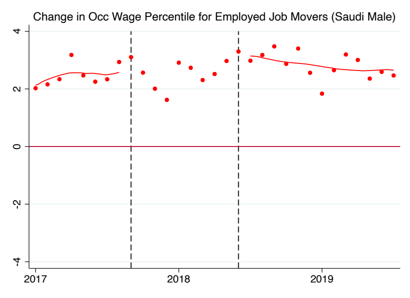




Occupation Improvement from Job Mobility

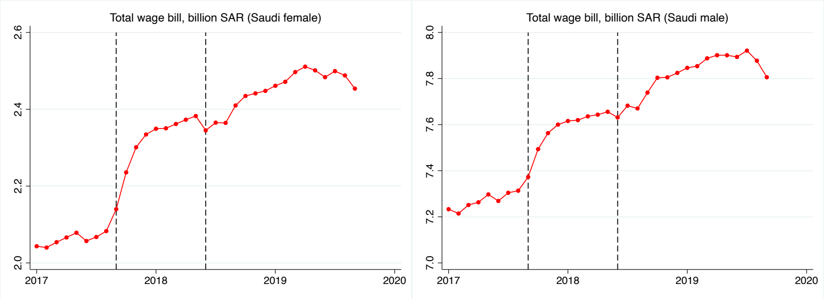
Job moves are also associated with transitions into higher-wage occupations.





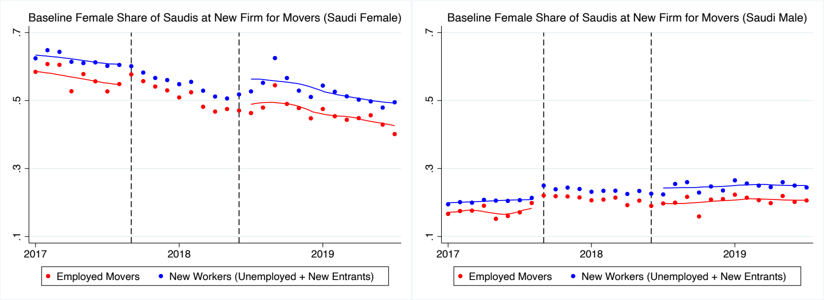
Wage Bill

Increased entry and stronger wage growth for employed workers yields significant increases in the total wage bill for women and men.



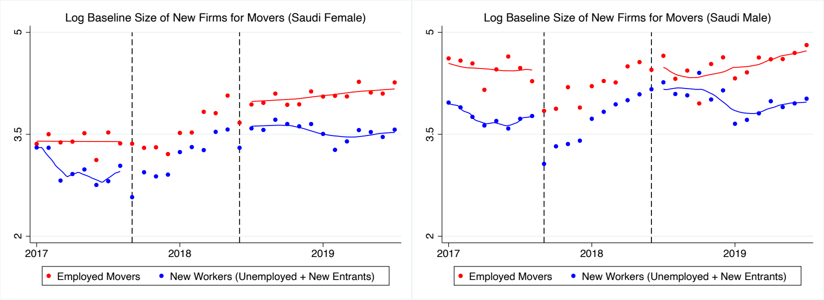
Worker Reallocation

Women join firms with (initially) fewer female employees after the implementation of the policy. Men are slightly more likely to join firms with more female employees.



Worker Reallocation

Women are also more likely to join larger firms after the lift of the ban.



Empirical Observations: Summary

The *announcement* of the change corresponds with:

- Increases in both women's and men's employment
- More even distribution of men and women across firms
- Increases in employment overall and within firms
- Increases in job mobility and entry into private sector work
- Consistent wage growth and occupation advancement from increased job mobility
- Increase in total wage bill for both men and women
- Movement of women to larger firms

Key Observations

- Workers and firms immediately adjust in anticipation of future policy change
- Lift announcement is associated with increases in the *number* and *quality* of job matches for women (and for men)
- Male and female workers reallocate across firms:
  - Previously all-male firms begin hiring women
  - Women move to firms with initially fewer female employees
  - Women move to larger firms

Model Overview

- Develop a simple model to account for salient empirical patterns:
  1. Anticipation of future policy changes drives decisions after announcement
  2. Extensive and intensive margin responses of women's employment (benefiting men as well)
  3. Improved allocation of workers (women and men) after implementation

- Three periods: baseline ($t = 0$), announcement ($t = 1$), and implementation ($t = 2$)

- Driving ban as a *wedge* between wage and marginal product (Hsieh et al. 2019, Chiplunkar and Goldberg 2024)

- Sunk cost of gender integration (Miller et al. 2022)
  - Extensive margin of women hiring; dynamics
  - Gender integration directly benefits women

- Imperfect substitution between women and men in production $\Rightarrow$ gender integration also improves productivity of men

Worker Preference and Labor Supply

- Discrete choice (Roy) model of firms

- Worker i with demographic (gender) $d_i \in \{m, f\}$ and idiosyncratic preference ε_{ijt}

- Utility from working for firm j:

$$U_{ijt} = \log \underbrace{W_{jt}(d_i)}_{\text{wage}} + \beta^{-1}\varepsilon_{ijt}$$

  - $\beta > 0$: strength of idiosyncratic preference
  - An outside option (home sector, indexed by 0) with wage $W_0(d_i)$

- Share of d workers of choosing firm j (with $\varepsilon \overset{iid}{\sim}$ Gumbel):

$$Pr_{jt}(d) = \frac{W_{jt}(d)^{\beta}}{W_0(d)^{\beta} + \sum_{k \in \mathcal{J}} W_{kt}(d)^{\beta}}$$

Firm Technology

- Firm j produces a homogeneous (numeraire) good using men (and potentially women)

- Efficiency unit of workers with demographic d:

$$N_{jt}(d) = \underbrace{\Phi_j(d)}_{\text{productivity}} \underbrace{(1 - \tau_t(d))}_{\text{wedge}} L_{jt}(d)$$

  - $\tau_t(d)$: time loss from commuting; $1 - \tau_t(d)$: effective labor supply
  - Assume $\tau_t(m) = 0$ and $\tau_t(f) > 0$

- Firm produces output combining efficiency units of men and women:

$$Y_{jt} = \left(N_{jt}(m)^{\frac{\sigma-1}{\sigma}} + \mathbb{I}_{jt} N_{jt}(f)^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \quad \sigma > 1$$

  - $\mathbb{I}_{jt}$: indicator for hiring women or not

Wage Setting and Profits

- Given $\square_{jt}$ and $\tau_t = \{\tau_t(m), \tau_t(f)\}$, firm j sets wage to maximize profits given labor supply

- (Log) Wage and profits:

$$\log W_{jt}(d) = \log \frac{\beta}{1+\beta} + \frac{\sigma-1}{\sigma+\beta} \underbrace{\log \Phi_j(d)}_{\text{productivity}} + \frac{\sigma-1}{\sigma+\beta} \underbrace{\log(1-\tau_t(d))}_{\text{wedge}} + \frac{1}{\sigma+\beta} \underbrace{\log X_{jt}(\square_{jt}, \tau_t)}_{\text{firm efficiency}} + \frac{1}{\sigma+\beta} \underbrace{\log \Lambda_t(d)}_{\text{competition}}$$

$$\Pi_{jt}(\square_{jt}, \tau_t) = \frac{1}{1+\beta} X_{jt}(\square_{jt}, \tau_t)$$

▸ Expressions for X and Λ

Wage Setting and Profits

- Partial equilibrium predictions, holding aggregate competition $\Lambda_t(d)$ fixed:

  1. $X_{jt}(1, \tau_t) > X_{jt}(0, \tau_t)$ when $\sigma > 1 \implies$ gender integration increases firm efficiency when men and women are imperfect substitutes ▸ Proof
  2. $\dfrac{\partial X_{jt}(1, \tau_t)}{\partial(1 - \tau_t(f))} > 0 \implies$ lower wedge for women increases efficiency of gender-integrated firms ▸ Proof

- Higher efficiency $\implies$ higher wage (and employment) for men and women, higher profits

Gender Integration

- Firms are endowed with $\Phi_j(m) > 0$ but $\Phi_j(f) = 0$
- Firms can pay a sunk cost S to draw productivity $\Phi_j(f) > 0$ and hire women $\mathbb{1}_{jt} = 1$
- State of firm j in period t: $(\Phi_j(f), \mathbb{1}_{jt}, \tau_t)$
  - Absorb $\Phi_j(m)$ in subscript
- Value of men-only firms:

$$V_{jt}(0,0,\tau_t) = \max_{\mathbb{1}_{jt}\in\{0,1\}} \mathbb{1}_{jt} \left\{-S + \mathbb{E}_{\Phi_j(f)}\left[\Pi_{jt}(\Phi_j(f),1,\tau_t) + V_{jt+1}(\Phi_j(f),1,\tau_{t+1})\right]\right\} \\ +(1-\mathbb{1}_{jt})\left\{\Pi_{jt}(0,0,\tau_t) + V_{jt+1}(0,0,\tau_{t+1})\right\}$$

- Value of gender-integrated firms:

$$V_{jt}(\Phi_j(f),1,\tau_t) = \Pi_{jt}(\Phi_j(f),1,\tau_t) + V_{jt+1}(\Phi_j(f),1,\tau_{t+1})$$

Model Predictions of Policy Effects

- Firms learn in period 1 that $\tau(f)$ will be lower in period 2
- Predicted effects after announcement:
  - Higher expected future profits from gender-integration induces more men-only firms to pay sunk costs to hire women
  - Gender-integration increases firm efficiency, which raises wage and employment for both men and women
- Predicted effects after implementation:
  - Lower $\tau(f)$ increases firm efficiency, which further raises wage and employment for both men and women at gender-integrated (likely larger) firms
  - Better labor market outcomes for women from the policy overall

Next Steps

- Potential extensions to the model:
  - Richer heterogeneity: skill, occupation, location
  - Varying amenities: policy may signal changes in social norms that affect women's taste for work/firms' taste for women for non-productive reasons
  - Worker dynamics: capture UE/EU/EE transitions for individual workers
- Estimate productivity and wedge to match data $\implies$ general equilibrium effects of policy on aggregate productivity/output
- Counterfactual exercises: how would the effects of policy change if, for example, the sunk costs of gender integration were lower?

Summary

- The announcement of the lift of the women's driving ban in Saudi Arabia led to increases in employment, worker reallocation, and improved match quality for female (and male) workers
- We model the ban as a wedge between the women's wages and marginal products, distorting demand for female workers
- Model matches key observations: firms and workers react to announcement (anticipating implementation), firms pay the fixed costs of gender integration, gender integration increases efficiency, and higher effiency yields higher wages and employment rates
- Use model to estimate overall effects of policy change on aggregate productivity and output
- Possible extensions to the model incorporating changes in social norms (wedge in worker preferences), richer heterogeneity, worker dynamics

Appendix

Firm Wage Setting Problem

- Firm j sets wage to maximize profits (revenue minus wage bill) given labor supply:

$$\max_{W_{jt}(m), W_{jt}(f)} \; Y_{jt} - W_{jt}(m)L_{jt}(m) - W_{jt}(f)L_{jt}(f)$$

  subject to

$$L_{jt}(d) = \frac{W_{jt}(d)^{\beta}}{W_0(d)^{\beta} + \sum_{k \in \mathcal{J}} W_{kt}(d)^{\beta}} \bar{L}(d)$$

- Optimal wage:

$$W_{jt}(d) = \frac{\beta}{1+\beta} \Phi_j(d)^{\frac{\sigma-1}{\sigma+\beta}} (1 - \tau_t(d))^{\frac{\sigma-1}{\sigma+\beta}} X_{jt}^{\frac{1}{\sigma+\beta}} \Lambda_t(d)^{\frac{1}{\sigma+\beta}}$$

  where

$$X_{jt} = \left(\left(\frac{\Phi_j(m)^{1+\beta}(1 - \tau_t(m))^{1+\beta}}{\Lambda_t(m)} \right)^{\frac{\sigma-1}{\sigma+\beta}} + \square_{jt} \left(\frac{\Phi_j(f)^{1+\beta}(1 - \tau_t(f))^{1+\beta}}{\Lambda_t(f)} \right)^{\frac{\sigma-1}{\sigma+\beta}} \right)^{\frac{\sigma+\beta}{\sigma-1}}$$

$$\Lambda_t(d) = \left(\left(\frac{\beta}{1+\beta} \right)^{\beta} \frac{\bar{L}(d)}{W_0(d)^{\beta} + \sum_{k \in \mathcal{J}} W_{kt}(d)^{\beta}} \right)^{-1}$$

Back

Gender Integration Increases Firm Efficiency

Given $\Phi_j(d) > 0$, $1 - \tau_t(d) > 0$, $\Lambda_t(d) > 0$, one can show that when $\sigma > 1$:

$$
\begin{aligned}
X_{jt}(1, \tau_t) &= \left(\left(\frac{\Phi_j(m)^{1+\beta}(1 - \tau_t(m))^{1+\beta}}{\Lambda_t(m)} \right)^{\frac{\sigma-1}{\sigma+\beta}} + \square_{jt} \left(\frac{\Phi_j(f)^{1+\beta}(1 - \tau_t(f))^{1+\beta}}{\Lambda_t(f)} \right)^{\frac{\sigma-1}{\sigma+\beta}} \right)^{\frac{\sigma+\beta}{\sigma-1}} \\
&> \left(\left(\frac{\Phi_j(m)^{1+\beta}(1 - \tau_t(m))^{1+\beta}}{\Lambda_t(m)} \right)^{\frac{\sigma-1}{\sigma+\beta}} \right)^{\frac{\sigma+\beta}{\sigma-1}} \\
&= X_{jt}(0, \tau_t)
\end{aligned}
$$

Back

Lower Wedge for Women Increases Efficiency of Gender-Integrated Firms

Given $\Phi_j(d) > 0$, $1 - \tau_t(d) > 0$, $\Lambda_t(d) > 0$, $\mathbb{I}_{jt} = 1$, one can show that:

$$\frac{\partial X_{jt}(1, \tau_t)}{\partial(1 - \tau_t(f))} = \left(\left(\frac{\Phi_j(m)^{1+\beta}(1 - \tau_t(m))^{1+\beta}}{\Lambda_t(m)} \right)^{\frac{\sigma-1}{\sigma+\beta}} + \mathbb{I}_{jt} \left(\frac{\Phi_j(f)^{1+\beta}(1 - \tau_t(f))^{1+\beta}}{\Lambda_t(f)} \right)^{\frac{\sigma-1}{\sigma+\beta}} \right)^{\frac{\sigma+\beta}{\sigma-1}-1}$$

$$\times \left(\frac{\Phi_j(f)^{1+\beta}(1 - \tau_t(f))^{1+\beta}}{\Lambda_t(f)} \right)^{\frac{\sigma-1}{\sigma+\beta}-1} (1+\beta) \frac{\Phi_j(f)^{1+\beta}(1 - \tau_t(f))^{\beta}}{\Lambda_t(f)}$$

$$> 0$$

Back